

CBSE Practice Worksheet

Class 6 | Mathematics | Difficulty: Hard | Set E

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. Find HCF of $(x^2 - 4)$ and $(x^2 - 4x + 4)$

- (A) $x - 2$
- (B) $x + 2$
- (C) $(x-2)^2$
- (D) $x^2 - 4$

2. Simplify: $(x^3 - 8)/(x - 2)$

- (A) $x^2 + 2x + 4$
- (B) $x^2 - 2x + 4$
- (C) $x^2 + 4$
- (D) $x^2 - 4$

3. Solve: $3(x-2)/4 = (2x+1)/3$

- (A) $x = 14$
- (B) $x = 22$
- (C) $x = 26$
- (D) $x = 30$

4. Find: $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3})$

- (A) 2
- (B) 8
- (C) 15
- (D) $\sqrt{15}$

5. Factorize: $x^2 - 5x - 14$

- (A) $(x-7)(x+2)$
- (B) $(x+7)(x-2)$
- (C) $(x-7)(x-2)$
- (D) $(x+7)(x+2)$

6. What is the cube root of -125?

- (A) 5
- (B) -5
- (C) ± 5
- (D) undefined

7. If $2\blacksquare = 3\blacksquare = 6\blacksquare$, which is true?

- (A) $1/z = 1/x + 1/y$
- (B) $z = x + y$
- (C) $z = xy$
- (D) $z = x - y$

8. Solve: $|2x - 5| = 9$

- (A) $x = 7$ or -2
- (B) $x = 7$ or 2
- (C) $x = -7$ or 2

(D) $x = -7$ or -2

9. If $x + 1/x = 5$, find $x^2 + 1/x^2$

(A) 21

(B) 23

(C) 25

(D) 27

10. If $a:b = 3:4$ and $b:c = 5:6$, find $a:c$

(A) 3:6

(B) 5:8

(C) 15:24

(D) 1:2

Section B: Fill in the Blanks (1 mark each)

11. If $x^2 - 6x + 8 = 0$, then $x = \underline{\hspace{1cm}}$ or $\underline{\hspace{1cm}}$

12. If $\log x = 2$, then $x = \underline{\hspace{1cm}}$

13. $(x + 1/x)^2 - (x - 1/x)^2 = \underline{\hspace{1cm}}$

14. $\text{HCF} \times \text{LCM}$ of 12 and 18 = $\underline{\hspace{1cm}}$

15. If $a + b + c = 0$, then $a^3 + b^3 + c^3 = \underline{\hspace{1cm}}$

16. If $x + y = 7$ and $xy = 12$, then $x^2 + y^2 = \underline{\hspace{1cm}}$

17. If $5^x = 125$, then $x = \underline{\hspace{1cm}}$

18. $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - \underline{\hspace{1cm}}$

Section C: Short Answer Questions (2 marks each)

19. Find the value of x : $3^{(2x-1)} = 27^{(x-2)}$

20. Solve: $\log_{10}(x - 1) + \log_{10}(x + 1) = 3$

21. Solve: $|x - 3| + |x + 2| = 7$

22. Simplify: $(x^2 - 16)/(x^2 + 4)$

23. If α and β are roots of $x^2 - 5x + 6 = 0$, find $\alpha^3 + \beta^3$

24. Simplify: $(2 + \sqrt{3})/(2 - \sqrt{3})$

25. Solve: $x^2 - 7x + 12 = 0$

--- End of Worksheet ---

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